# Stochastic Attention via Langevin Dynamics on the Modern Hopfield Energy

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## Motivation

Attention heads are deterministic: the same query always returns the same softmax-weighted memory average. Yet many tasks require *generating* novel, plausible patterns from structured memory. Score-based diffusion models solve this for large datasets by learning a neural score function, but they require thousands of training examples and expensive optimization. When data is scarce, there is nothing to train on.

We observe that the gradient of the **modern Hopfield energy** is exactly the identity minus the attention map. Applying **Langevin dynamics** to this energy yields *stochastic attention*: a **training-free sampler** controlled by a single inverse temperature $\beta$. The score function is the attention map itself – no score network, no training loop, no contrastive objective.

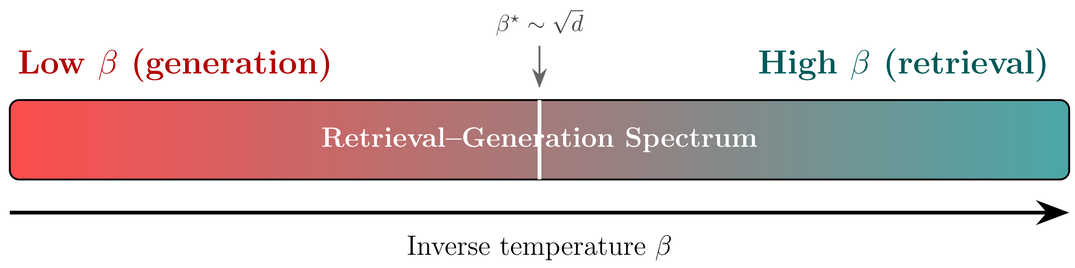


**Figure 1.** The retrieval–generation spectrum. A single inverse temperature $\beta$ controls sample behavior: low $\beta$ yields broad generation, high $\beta$ yields exact retrieval. The critical temperature $\beta^\star \sim \sqrt{d}$ marks the boundary, identified analytically via an entropy inflection condition.

A single parameter $\beta$ interpolates continuously between open-ended generation ($\beta \to 0$) and exact retrieval ($\beta \to \infty$). The critical temperature $\beta^\star$, identified analytically via an entropy inflection condition ($\beta^\star \sim \sqrt{d}$ for random unit-norm memories), marks the boundary between these two regimes.

## Method: Stochastic Attention Update

$\mathbf{X} = [\mathbf{m}_1, \ldots, \mathbf{m}_K] \in \mathbb{R}^{N \times K}$ stores $K$ patterns. To generate, we sample from $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta E(\boldsymbol{\xi}))$. The ULA turns any smooth energy $U$ into a sampler:

$$\boldsymbol{\xi}_{t+1} = \boldsymbol{\xi}_t - \tilde{\alpha}\, \nabla U(\boldsymbol{\xi}_t) + \sqrt{2\tilde{\alpha}}\, \boldsymbol{\epsilon}_t, \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

The gradient of the modern Hopfield energy is exactly the attention residual:

$$\nabla E(\boldsymbol{\xi}) = \boldsymbol{\xi} - \underbrace{\mathbf{X}\,\mathrm{softmax}(\beta\, \mathbf{X}^\top \boldsymbol{\xi})}_{\mathbf{T}(\boldsymbol{\xi}) \,=\, \text{softmax attention map}}.$$

Substituting $\nabla E$ into the ULA (with $U = \beta E$, $\tilde{\alpha} = \alpha/\beta$):

$$\boldsymbol{\xi}_{t+1} = \underbrace{(1-\alpha)\,\boldsymbol{\xi}_t}_{\text{contraction}} + \underbrace{\alpha\, \mathbf{X}\,\mathrm{softmax}(\beta\, \mathbf{X}^\top \boldsymbol{\xi}_t)}_{\text{attention pull to memories}} + \underbrace{\sqrt{\frac{2\alpha}{\beta}}\, \boldsymbol{\epsilon}_t}_{\text{Gaussian noise}}$$

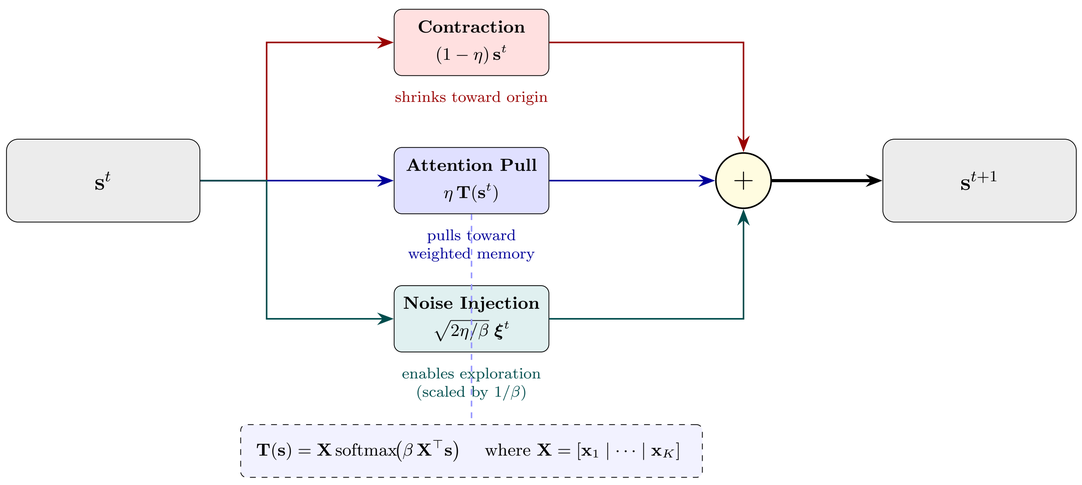


**Figure 2.** The stochastic attention update decomposes into three interpretable components: contraction toward the origin, an attention pull toward stored memories, and isotropic Gaussian noise scaled by temperature. Each step costs $\mathcal{O}(dK)$, identical to one attention head, with no training required. Convergence is $W_2$-geometric in the convex regime ($\beta\sigma_{\max}^2 < 2$); MALA acceptance = 99.2% at $\alpha = 0.01$ confirms negligible discretization bias. The critical temperature $\beta^*$ is selected via the entropy inflection condition $d^2H/d\beta^2 = 0$ and scales as $\beta^* \sim \sqrt{d}$.

## Experiment 1:Phase Behavior & Convergence (Synthetic)

The theory predicts that $\beta$ acts as a continuous order parameter, with a smooth transition between generation and retrieval centered near $\beta^* \sim \sqrt{d}$. We test this on synthetic random unit-norm patterns and verify convergence to the correct Boltzmann target.

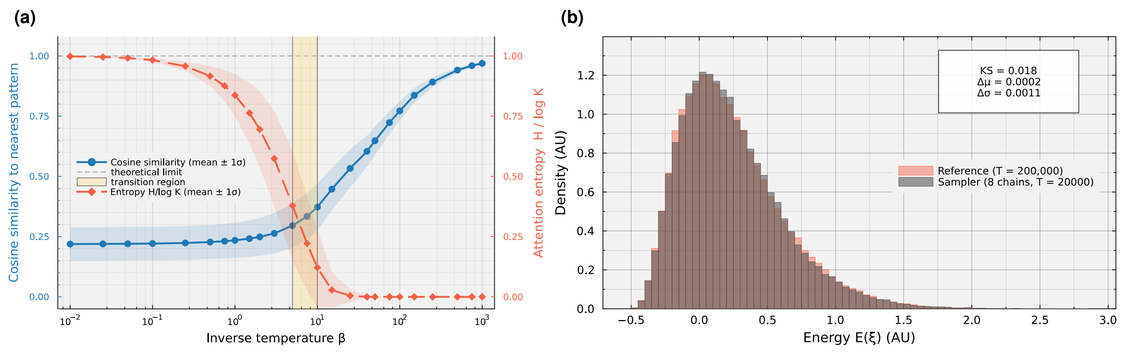


**Figure 3. (a)** $d = 64$, $K = 16$. Both cosine similarity to the nearest pattern (blue) and attention entropy (coral) undergo a smooth sigmoidal transition centered at $\beta \approx 5$–10 (gold band), consistent with the predicted $\beta^* \sim \sqrt{d} \approx 8$. **(b)** $d = 8$, $K = 4$, $\beta = 5$. Eight independent chains (gray) converge to a long-run reference distribution (coral); KS testing confirms the sampler reaches the correct Boltzmann target.

## References

[1] Ramsauer et al., "Hopfield Networks is All You Need," *ICLR*, 2021.
[2] Roberts & Tweedie, "Exponential Convergence of Langevin Distributions and Their Discrete Approximations," *Bernoulli*, 2(4):341–363, 1996.
[3] Durmus & Moulines, "Nonasymptotic Convergence Analysis for the Unadjusted Langevin Algorithm," *Ann. Appl. Probab.*, 27(3):1551–1587, 2017.
[4] Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models," *NeurIPS*, 33:6840–6851, 2020.

## Experiment 2:MNIST Image Generation ($K = 100$, $d = 784$)

With the phase behavior confirmed, we move to structured data. We store $K = 100$ images of digit "3" and compare stochastic attention against both synthetic baselines (bootstrap, Gaussian perturbation, convex combinations, GMM-PCA) and learned generative models (VAE, DDPM). We run 30 independent chains at $\beta = 2000$ (retrieval) and $\beta = 200$ (generation, near $\beta^*$), collecting 150 samples per regime.

| Method | Nov.↑ | Div.↑ | Energy↓ |
|---|---|---|---|
| Bootstrap | 0.000 | 0.459 | −0.500 |
| Gaussian perturb. | 0.004 | 0.450 | −0.496 |
| Convex combo | 0.092 | 0.008 | −0.399 |
| GMM-PCA | 0.198 | 0.419 | −0.303 |
| VAE (latent = 8) | 0.214 | 0.441 | −0.286 |
| DDPM† | 0.938 | 0.991 | +0.44 |
| MALA ($\beta$=2000) | 0.151 | 0.598 | −0.305 |
| SA ($\beta$=2000) | 0.152 | 0.600 | −0.303 |
| **SA ($\beta$=200)** | **0.548** | **0.885** | +1.467 |

†DDPM generates isotropic noise despite 5,000 training epochs.

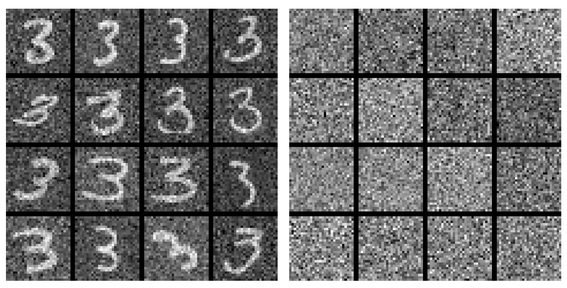


**Figure 4. (Left)** Novelty, diversity, and energy scores for all methods on MNIST digit "3" ($K$=100, $d$=784). SA at generation temperature ($\beta$=200) achieves the highest novelty and diversity among energy-based methods. **(Right)** Sample grids from SA ($\beta$=2000, retrieval regime) and DDPM (5,000 training epochs). DDPM produces isotropic noise on this small dataset.

In the generation regime, SA achieves **2.6**× the novelty and **2.0**× the diversity of the best learned baseline (VAE), while matching the Metropolis-corrected MALA gold standard. DDPM fails entirely on this small dataset, producing isotropic noise despite 5,000 training epochs. This illustrates the core advantage of a training-free score: when data is scarce, there is no distribution to learn, but the exact Hopfield gradient is always available.

## Experiment 3:Protein Sequence Generation (Pfam RRM, $K = 68$, $d = 59$)

Protein sequences pose a harder test than images: generated samples must preserve not just global amino acid composition but position-specific conservation and pairwise co-evolutionary couplings that define family membership. We store 68 aligned RRM (RNA Recognition Motif) sequences from Pfam, one-hot encoded and PCA-reduced to $d$=59 (95% variance retained). The entropy inflection gives $\beta^*$=3.85, half the random-pattern prediction $\sqrt{d}$=7.68, because conserved residues concentrate the similarity spectrum.

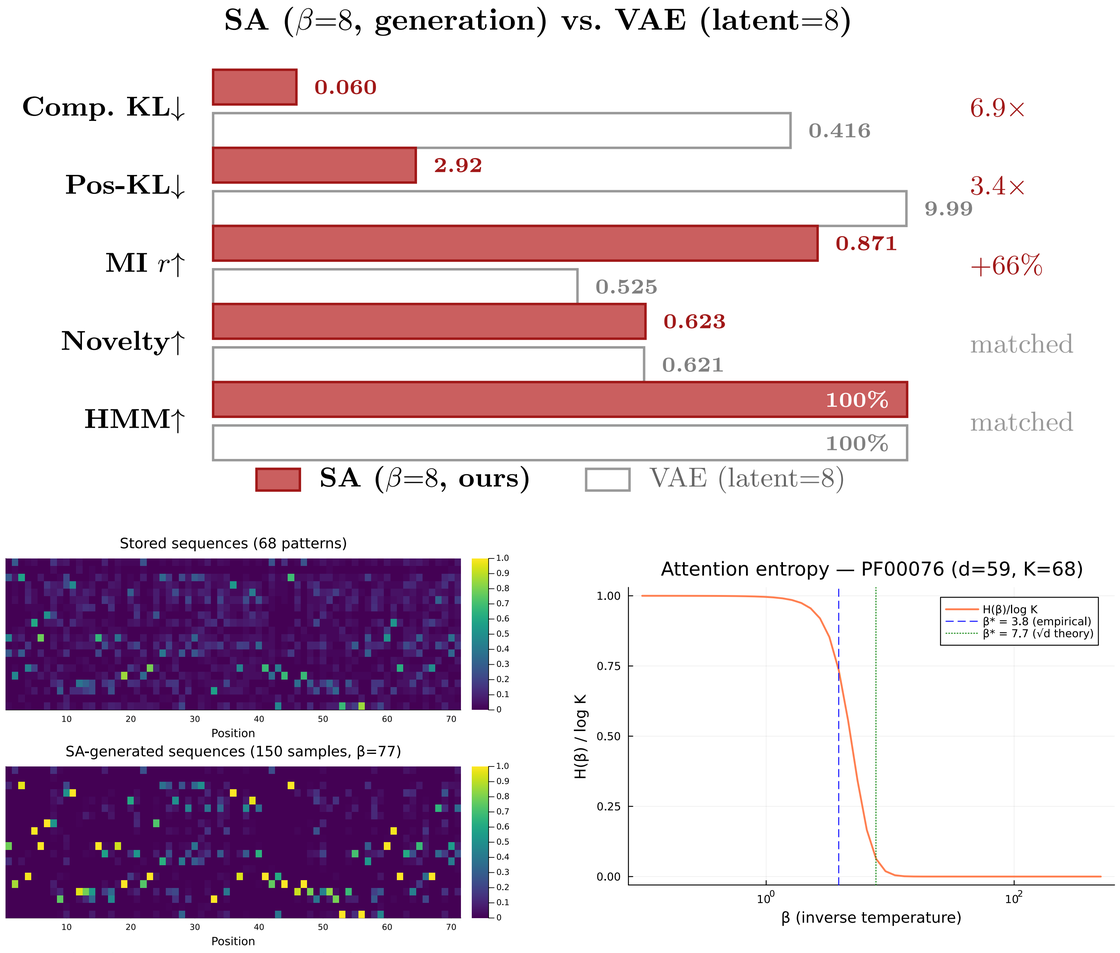


**Figure 5. (Top)** Composition divergence (KL), per-position divergence, and coupling fidelity for SA vs. VAE on the Pfam RRM family ($K$=68, $d$=59). **(Bottom-left)** Stored (top) vs. SA-generated (bottom) amino acid frequencies. **(Bottom-right)** Entropy inflection identifies $\beta^*$=3.85, the retrieval–generation boundary.

At generation temperature, SA matches the VAE on novelty while achieving **6.9**× lower composition divergence, **3.4**× lower per-position divergence, and **66**% higher coupling fidelity. Every generated sequence passes the Pfam HMM filter (median $E$-value $8.1 \times 10^{-34}$). Because the Langevin gradient is recomputed from stored sequences at every step, it continuously steers samples toward the family's structural profile – fidelity that the VAE, trained on only 68 examples, cannot maintain.

## Key Result

Stochastic attention is a **training-free** generative sampler whose score function is the attention map itself. Each step costs $\mathcal{O}(dK)$ – the same as one attention head – with no learned parameters, no training loop, and no dataset size requirement. A single analytically identified parameter $\beta^*$ controls the generation–retrieval tradeoff.

On **MNIST** ($K$=100), SA achieves **2.6**× the novelty and **2.0**× the diversity of the best learned baseline (VAE), while DDPM fails entirely. On **protein sequences** ($K$=68), SA achieves **6.9**× lower composition divergence and **66**% higher coupling fidelity than the VAE, with 100% Pfam HMM pass rate.